6COM2012: Social and Collective Artificial Intelligence
Worksheet 1:
Co-existing Agents

This worksheet focuses on interactions that occur between agents simply by co-existing in a shared world.

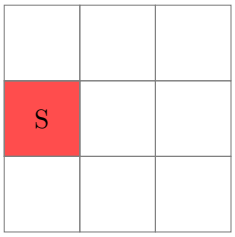
1. **Basic interactions**

   Implement a 3 by 3 grid world with 3 agents in Python.
   If an agent leaves the world on one side, it re-enters on the other side again. Each agent should have its own behaviour and moves in a specific pattern:

   - Walk clockwise around the world.
   - Walk diagonally in a diamond shape counter-clockwise.
   - Walk always to the left.

   If an agent tries to move to a field occupied by another agent, the move is cancelled.

   Each agent starts in the red field but two time-steps apart. Define the order in which the agents move each round. Choose the order in which the agents start.



   Run the simulation for at least 50 time steps.
   Each time there is a conflict, print the field and agents involved. Do you see a pattern?

2. **Platform Royal**

   Create a 5 by 5 grid world with no walls around it. Any agent that moves off the grid permanently leaves the game.
   This time, if an agent tries to move to a field occupied by another agent, it pushes the other agent by one field.
   Add at least 4 different agents with individual movement patterns and defined start states to this world. Define the order in which the agents move each round.
   Run the simulation 3 times with different start states for the agents and record how many agents remain in the world after 100 time-steps.

   Challenge (optional): Design your own agent that pushes all other agents from the platform. You can achieve this by clever design an incidental antagonism or by giving the agent new capabilities making it intentionally antagonistic.